

Design of experiments in PHARMA

From model-based clinical studies to structured clinical trials

mODa 14

V. V. Fedorov

Standard steps in optimal design

Traditional model-based design of experiments, gaining maximum information per experimental unit

- Model: $Y \sim p(y|\eta) \ , \quad \eta = \eta(x, \theta)$
- Estimation method: *Maximum likelihood*
- Design region/controls: $x \in \mathcal{X} \quad \xi = \{w_i, x_i\}_1^n$
- Criteria of optimality: $\Psi [M(\xi, \theta)] \ , \quad M(\theta, \xi) = \sum_{i=1}^n w_i \mu(x_i, \theta)$
- Optimization problem: $\xi^* = \arg \min_{\xi \in \Xi} \Psi [M(\xi, \theta)]$

Notations

- Elemental information matrix:

$$\nu(\eta) = \text{Var} \left[\frac{\partial}{\partial \eta} \ln p(y|\eta) \right]$$

See the complementary slides for tables containing eFIM for the most popular distributions.

- Information matrix of a single observation:

$$\mu(x, \theta) = F(x, \theta) \nu(\eta) F^T(x, \theta),$$

$$F(x, \theta) = \frac{\partial \eta^T(x, \theta)}{\partial \theta}$$

- Asymptotic variance – covariance matrix

$$D = M^{-1}$$

Table of popular criteria and sensitivity functions

$\Psi(\xi)$	$\psi(x, \xi)$
$\log D , \quad D ^{1/m}, \quad \prod_{\alpha=1}^m \lambda_{\alpha}(D)$	$\operatorname{tr} \nu(\eta) F^T(x) D F(x) - m$
$\log A^T D A ,$ $\dim A = k \times m, \operatorname{rank} A = k < m$	$\operatorname{tr} \nu(\eta) F^T(x) D A (A^T D A)^{-1} A^T D F(x) - k$
$\operatorname{tr} D, \quad \sum_{\alpha=1}^m \lambda_{\alpha}(D)$	$\operatorname{tr} \nu(\eta) F^T(x) D^2 F(x) - \operatorname{tr} D$
$\operatorname{tr} A^T D A$	$\operatorname{tr} \nu(\eta) (F^T(x) D A)^2 - \operatorname{tr} A^T D A$
$\operatorname{tr} D^{\gamma}, \quad \sum_{\alpha=1}^m \lambda_{\alpha}^{\gamma}(D)$	$\operatorname{tr} \nu(\eta) F^T(x) D^{\gamma+1} F(x) - \operatorname{tr} D^{\gamma}$
$\lambda_{\min} = \lambda_{\min}(M)$ $= \lambda_{\max}^{-1}(D)$	$\sum_{i=1}^a \pi_i P_i^T F(x) \nu(\eta) F(x) P_i - \lambda_{\min}$ $\lambda_{\min} P_i = M P_i,$ a is the multiplicity of $\lambda_{\min},$ $\sum_{i=1}^a \pi_i = 1, 0 \leq \pi_i \leq 1$

Exploratory phases

Model-based design and data analysis

Phase 0

This is an exploratory phase involving a very small number of participants (usually fewer than 15). It aims to gather preliminary data on how the drug behaves in the human body using subtherapeutic doses.

Helps researchers decide whether to proceed with further clinical development based on early pharmacokinetic data

Phase I

Conducted with 20 to 80 healthy volunteers, this phase focuses on assessing safety, tolerability, and optimal dose. Participants are closely monitored for side effects, and the best method of administration (oral, intravenous, etc) is determined.

Approximately 70% of drugs move on to Phase II.

Phase II

Involves several hundred participants who have the condition the drug aims to treat. This phase evaluates the drug's effectiveness and further assesses its safety. Participants receive the dosage found to be safe in Phase I, and researchers monitor them for several months to gather data on efficacy and side effects. About 33% of drugs progress to Phase III from here

Multiple responses/indicators problem

Toxicity	Typical study	
	Placebo	Treatment
Any SAE	30%	70%
Diarrhea	0	8%
Cardiac disorders	2%	2%
Nausea/Vomiting	5% / 3%	10% / 10%
Anemia	3%	5%
Hepatotoxicity	3%	10%
Blood and lymphatic system	4%	6%
Gastrointestinal	2%	4%
Hypertension	3%	12%
Metabolism and nutrition disorders	5%	8%

Y = 1, if there is any SAE, Y = 0 otherwise

Example 1: Binary dose-response model

		Toxicity Z		
		1	0	
Efficacy Y	1	p_{11}	p_{10}	$p_{1\bullet}$
	0	p_{01}	p_{00}	$p_{0\bullet}$
		$p_{\bullet 1}$	$p_{\bullet 0}$	

Probit model

$$p_{00}(x, \boldsymbol{\theta}) = 1 - \Phi[\eta_1(x, \boldsymbol{\theta}), \eta_2(x, \boldsymbol{\theta}); \rho] ,$$

$$p_{0\bullet} = \Phi[\eta_1(x, \boldsymbol{\theta})] ,$$

$$p_{\bullet 0} = \Phi[\eta_2(x, \boldsymbol{\theta})] ,$$

$$p_{10} = 1 - p_{\bullet 1} - p_{00}, \quad p_{01} = 1 - p_{1\bullet} - p_{00}$$

$$\Phi[\eta_1, \eta_2; \rho] = \int_{-\infty}^{\eta_1} \int_{-\infty}^{\eta_2} \frac{1}{2\pi\sqrt{1-\rho^2}} e^{-\frac{z_1^2 - 2\rho z_1 z_2 + z_2^2}{2(1-\rho^2)}} dz_1 dz_2 ,$$

$$\Phi[\eta] = \int_{-\infty}^{\eta} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz .$$

Information matrix for a single observation

$$\mu(x, \theta) = \frac{\partial \mathbf{p}}{\partial \theta} (\mathbf{P} - \mathbf{p} \mathbf{p}^\top)^{-1} \frac{\partial \mathbf{p}}{\partial \theta^\top},$$

$$\mathbf{p} = \begin{pmatrix} p_{11} \\ p_{10} \\ p_{01} \end{pmatrix} \text{ and } \mathbf{P} = \begin{pmatrix} p_{11} & 0 & 0 \\ 0 & p_{10} & 0 \\ 0 & 0 & p_{01} \end{pmatrix},$$

or in common words:

*Information gained at point x =
Squared rate of response change/Response variance*

To learn or to treat?

Penalty function $\phi(x)$:

$$\text{Penalty} = \{ \text{Probability of efficacy without toxicity} \}^{-C_E} \{ 1 - \text{Probability of toxicity} \}^{-C_T}.$$

Design region :

$$x \in \mathcal{X} = \{ \text{Probability of efficacy} > Q_E \text{ and Probability of toxicity} < Q_T \}.$$

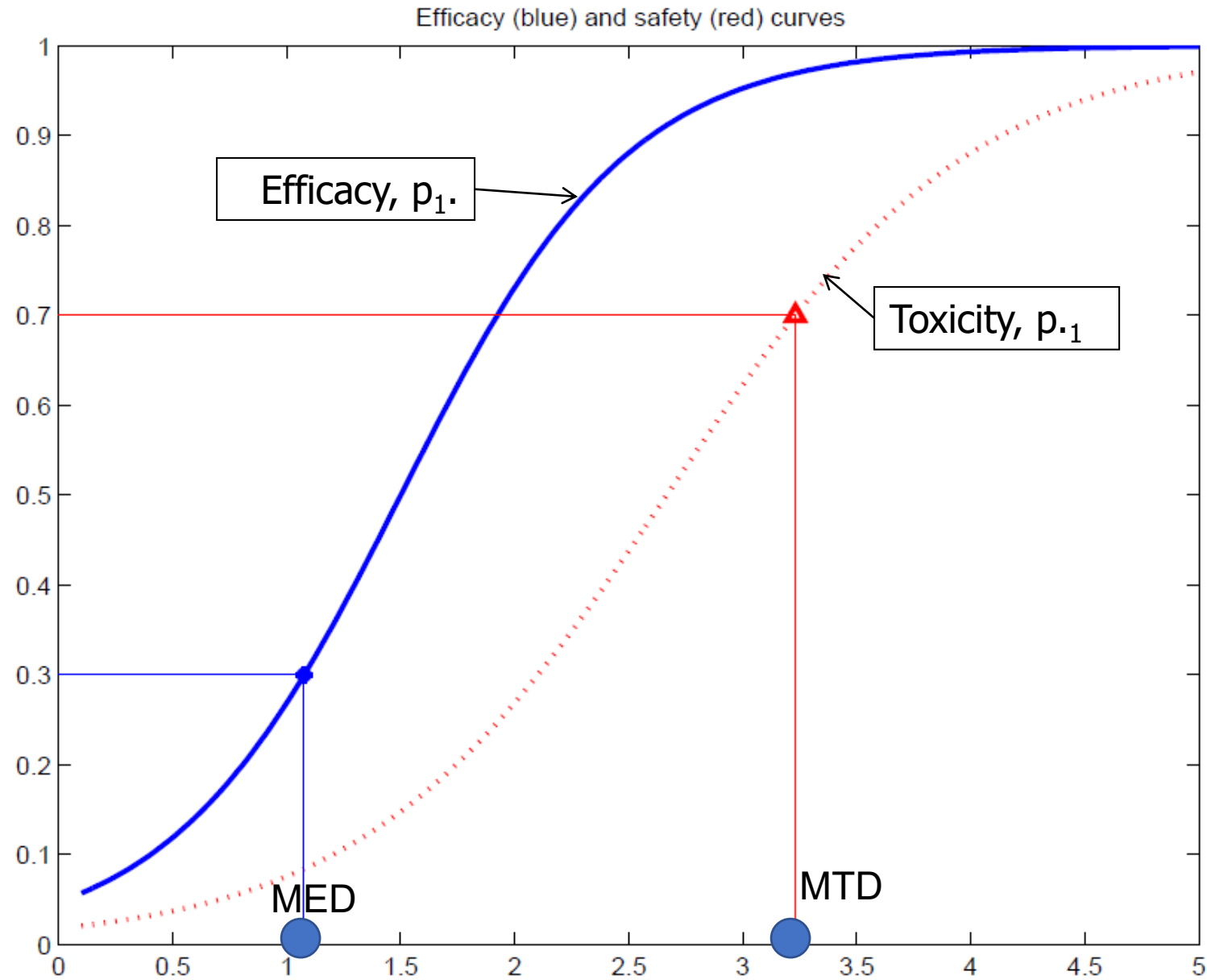
To learn:

- Increase C_E and C_T
- Decrease Q_E and increase Q_T

To treat:

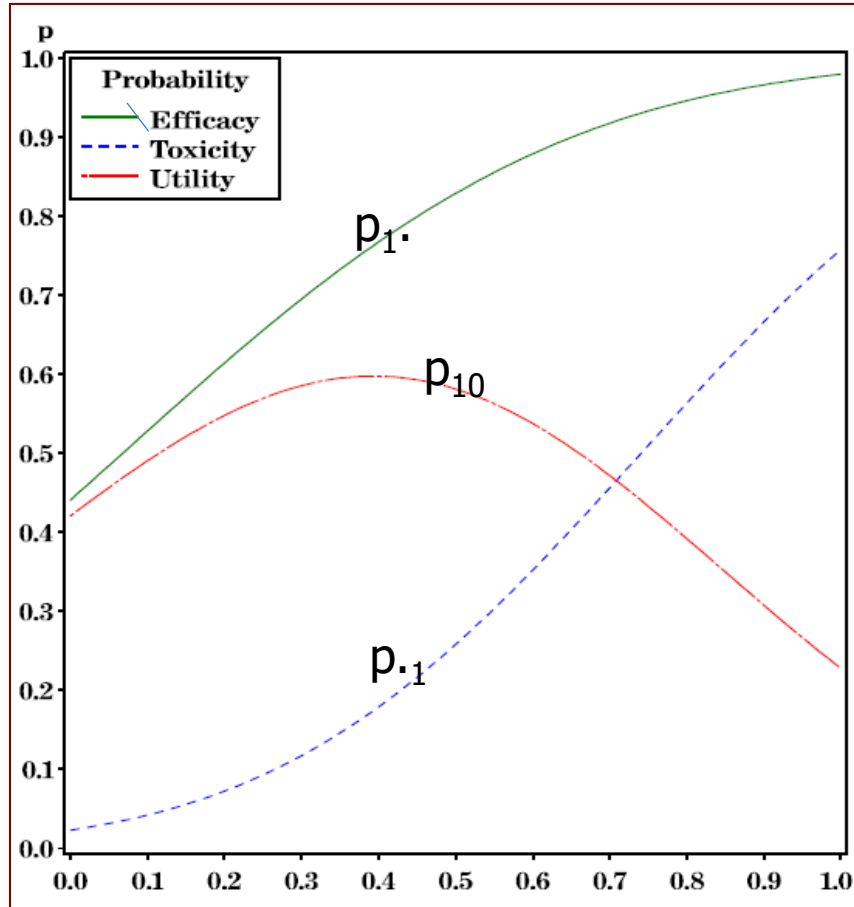
- Decrease C_E and C_T
- Increase Q_E and decrease Q_T

Adding safety concerns

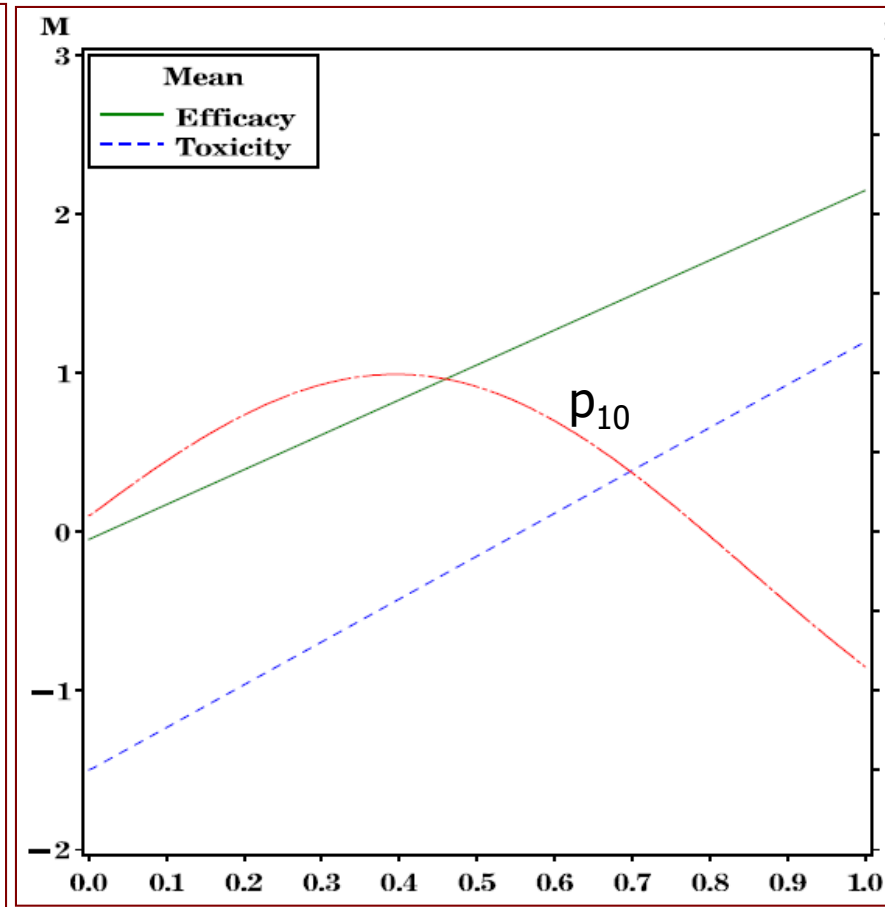


Typical dose-response curves

Probit model



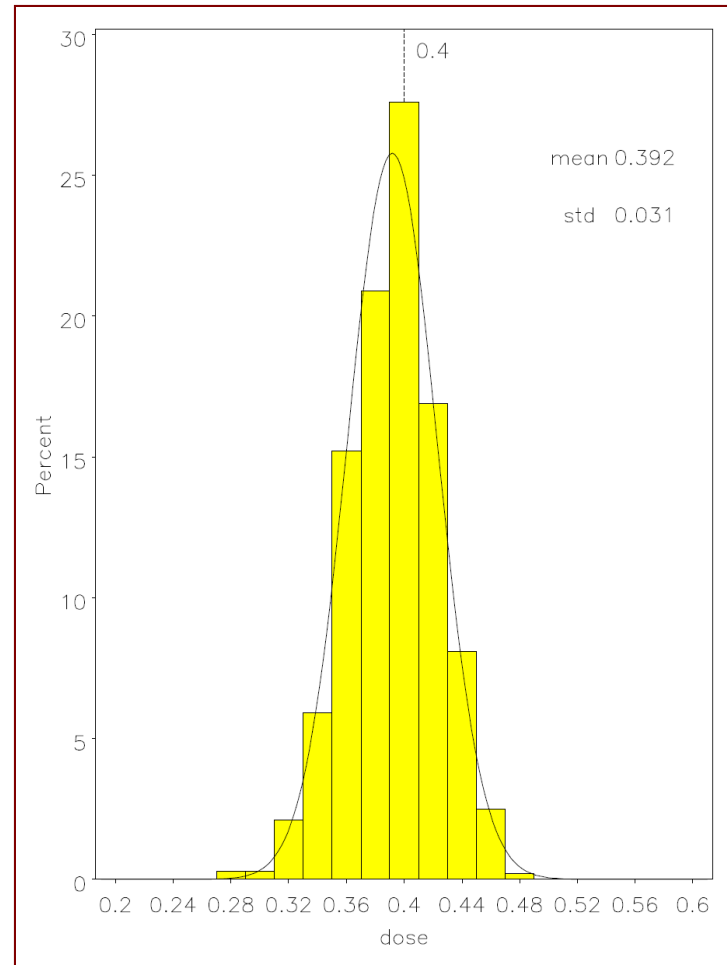
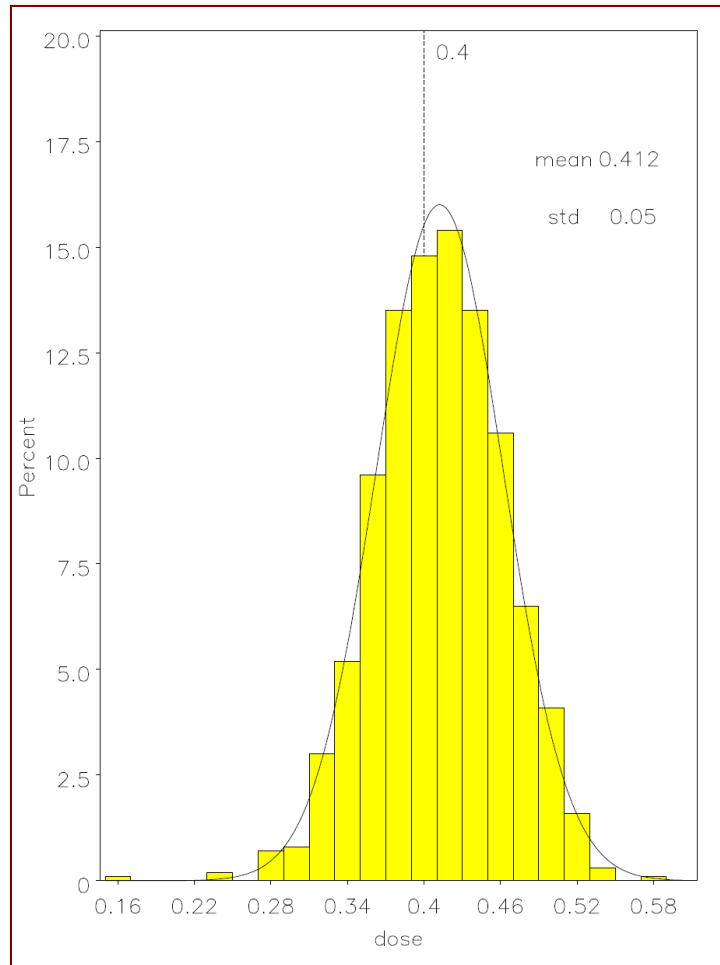
Responses and reported end-point
are dichotomized



Responses are continuous, reported end-
point is dichotomized

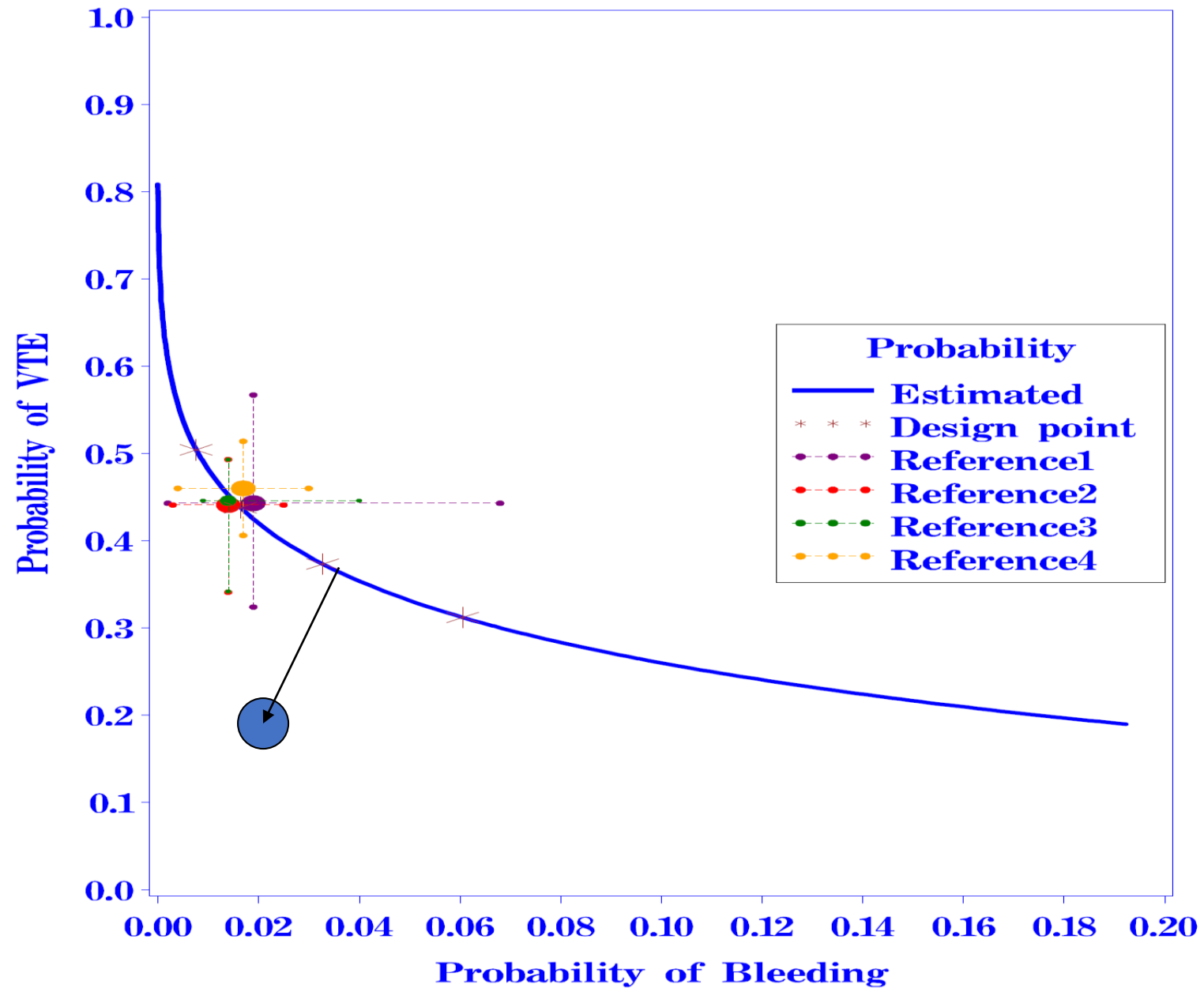
Model

Estimation of the best dose with and without dichotomization

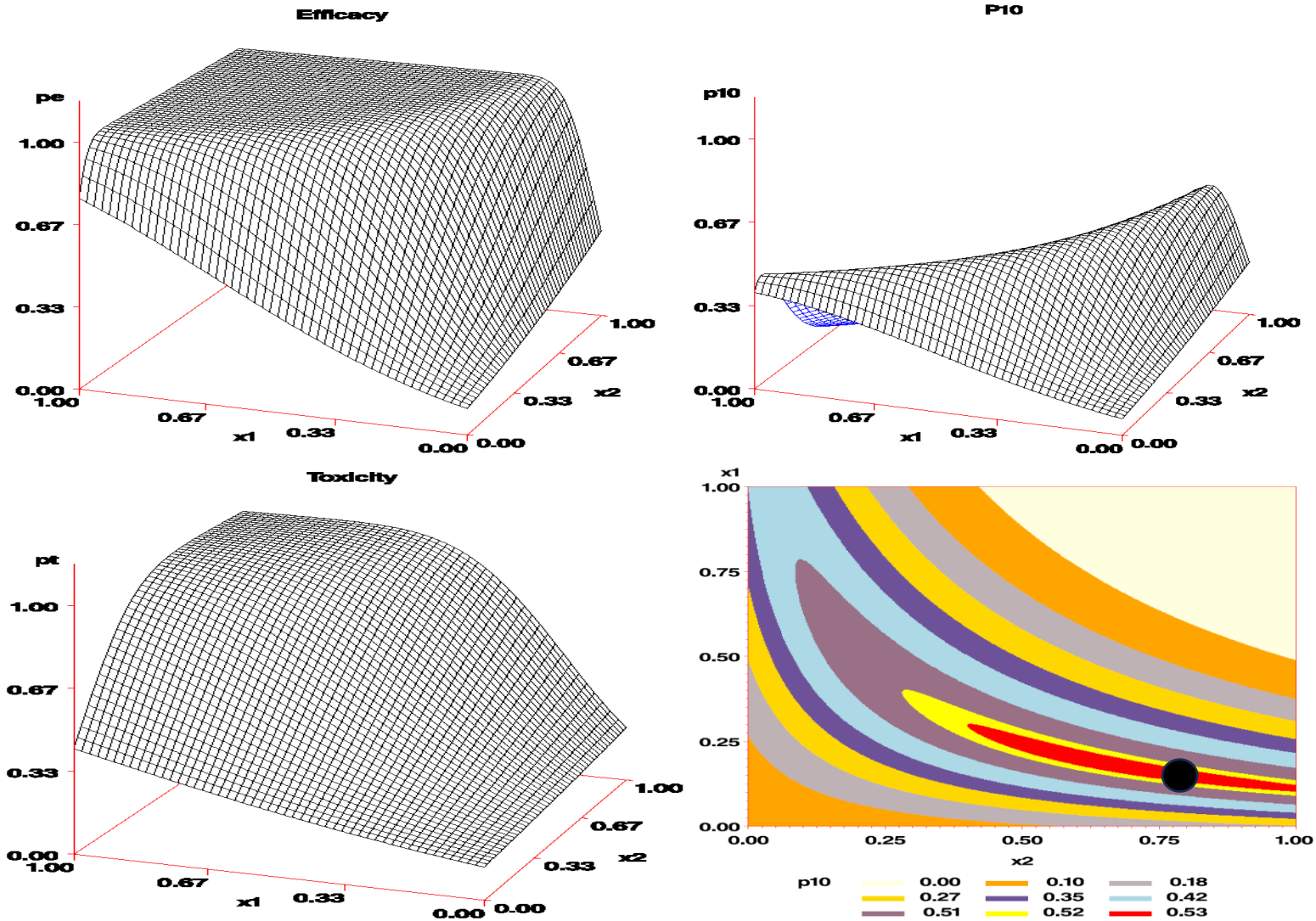


Simulation results for two stage D-optimal designs

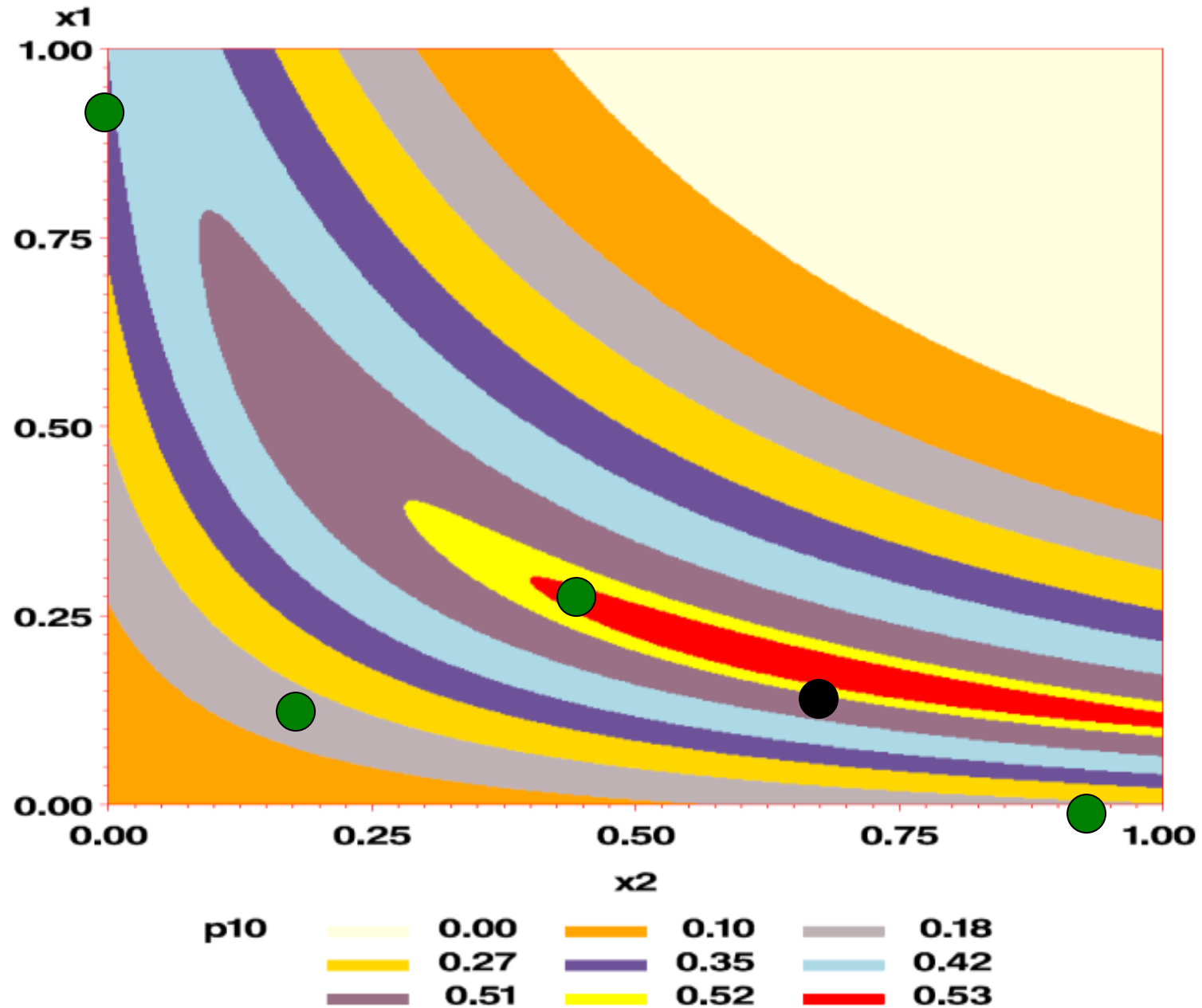
Step towards the individual dose tuning



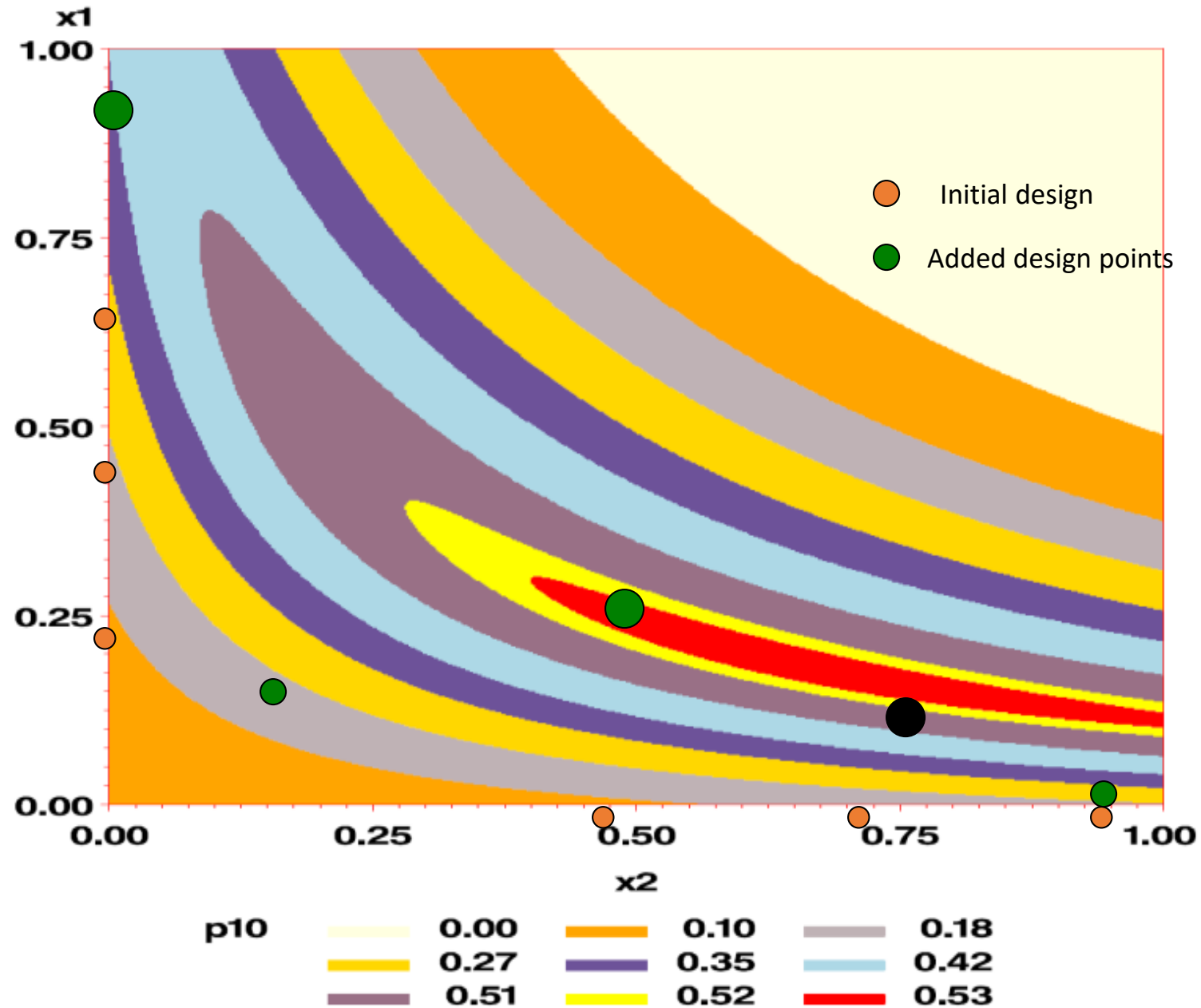
Example 2. Drug combinations



Penalized locally D-optimal



Penalized composite locally D-optimal



Design comparison for D-criterion, including initial design

Design type	Information per patient	Penalty per patient	Information per penalty u.
Locally D-optimal	1	2.07	0.48
Penalized locally optimal	0.79	1	0.79
Penalized composite locally optimal	0.77	1.23	0.62
Penalized composite (median, 500)	0.70	1.36	0.52
Penalized adaptive (median, 500)	0.75	1.50	0.50
"Do your best" adaptive (median, 500)	0.46	1.19	0.39
Restricted "Uniform " , 12 points	0.67	1.34	0.50

Later phases

Phase II

Involves several hundred participants who have the condition the drug aims to treat. This phase evaluates the drug's safety and effectiveness. Participants receive the dosage found to be safe in Phase I, and researchers monitor them for several months to gather data on efficacy and side effects. About 30% of drugs progress to Phase III

Phase III

This phase tests the drug on a larger population (hundreds to thousands) to confirm its effectiveness, monitor side effects, and compare it to standard or equivalent treatments. Successful completion of Phase III is required for regulatory approval. It provides comprehensive data on the drug's performance in a real-world setting. Successful outcome about 60%.

Phase IV

Known as post-marketing surveillance, this phase occurs after the drug has been approved for public use. It monitors long-term effectiveness and safety in a larger population. Ongoing studies monitor rare side effects or long-term effects that were not apparent in earlier phases

Major players



Regulatory authority, shareholders, etc.

Example 3. Dose tuning

Model:

$$E[y|x] = \eta(x, \theta)$$

Type I:

$$x^*(\theta) = \text{Arg}[\eta(x, \theta) = C]$$

Type II:

$$x^*(\theta) = \text{Arg} \max_x \eta(x, \theta)$$

Losses of major players

- Targeted population
- Sampled patients
- Specific patient
- Sponsor

$$\sim L_{po} = (\hat{x}_N - x^*)^2$$

$$\sim L_{sa} = \sum_{n=1}^N (x_{n+1} - x^*)^2$$

$$\sim L_{pa} = (x_{n+1} - x^*)^2$$

$$\sim L_{sp} = N$$

Best intention vs most informative

- Best intention designs:

Allocate every new patient to the BEST (accordingly to the current knowledge) dose

$$x_{n+1} = x^*(\hat{\theta}_n) = \text{Arg max}_x \eta(x, \hat{\theta}_n)$$

- Most informative designs:

Allocate every new patient to the dose that is MOST INFORMATIVE

$$x_{n+1} = \text{Arg max}_{x \in \mathbf{X}} \{\text{Var}[\eta(x, \hat{\theta}_n)]\}$$

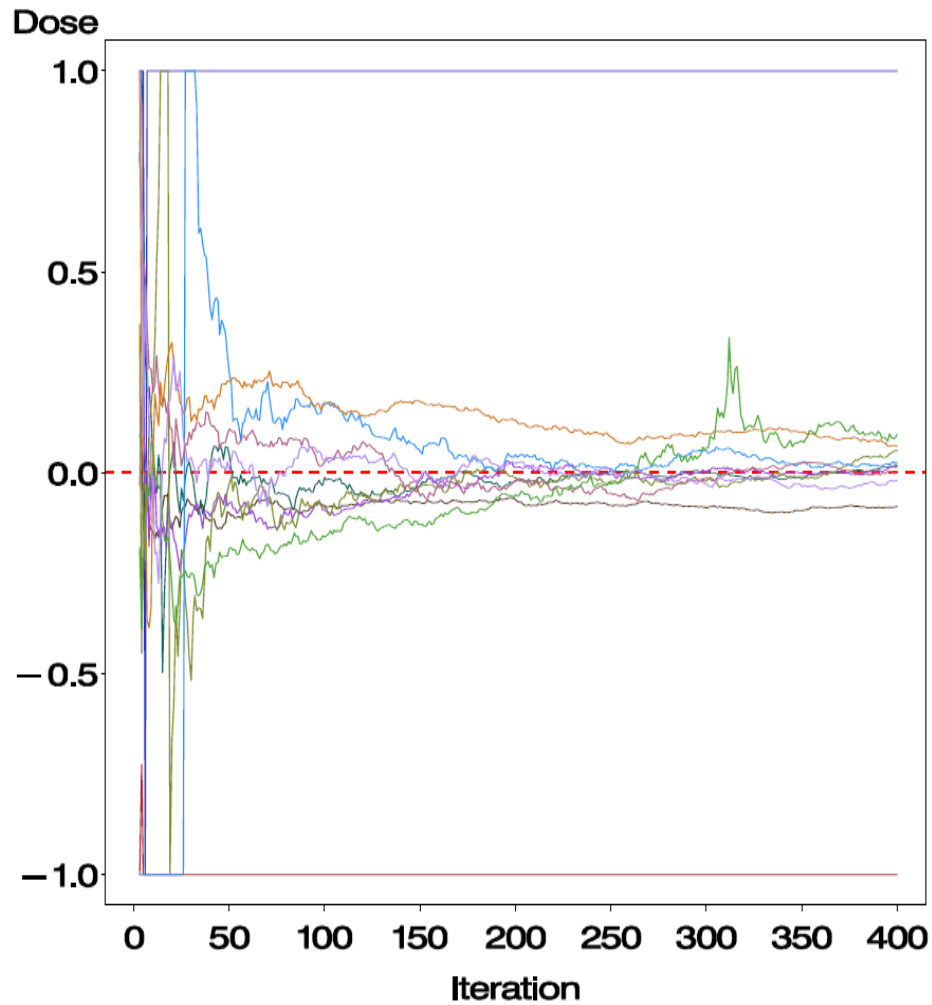
- Compromise:

Use PENALIZED adaptive designs

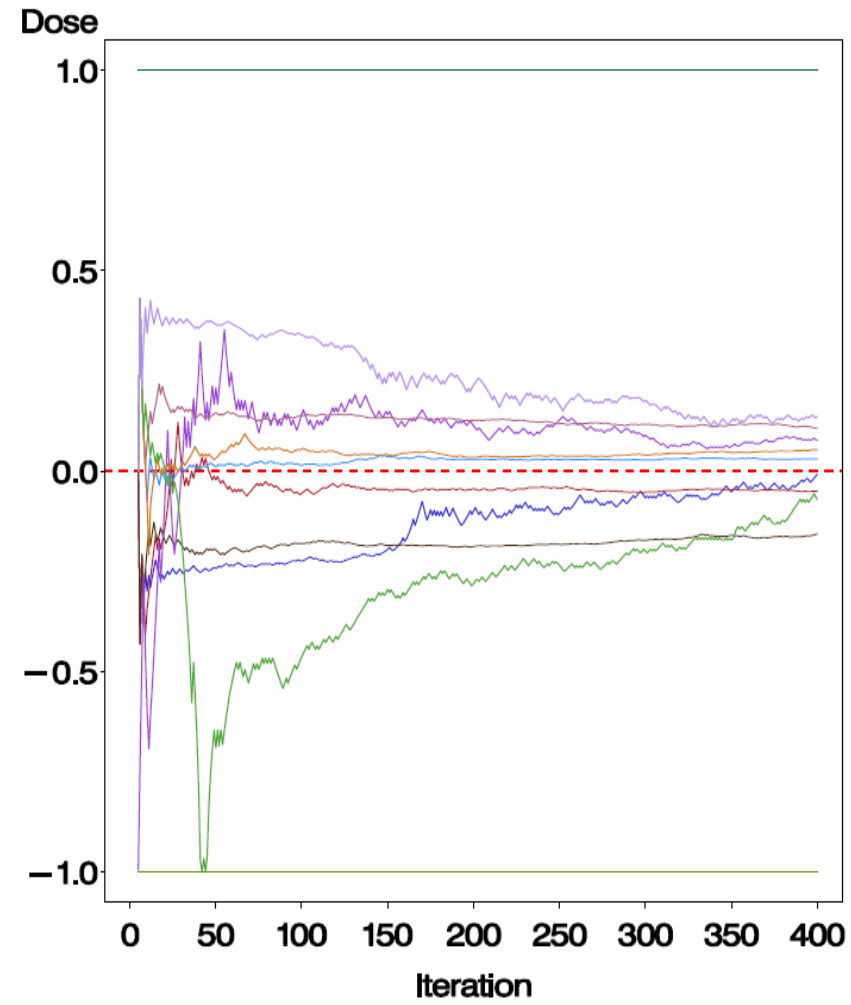
$$x_{n+1} = \text{Arg max}_{x \in \mathbf{X}} \{\text{Var}[\eta(x, \hat{\theta}_n)] - w_n[\eta_n^* - \eta(x, \hat{\theta}_n)]^2\}$$

$$\eta_n^* = \max_{x \in \mathbf{X}} \eta(x, \hat{\theta}_n)$$

A few BID trajectories for the Type I problem



Linear regression



Logistic binary (with linear link)

Response optimization: self-tuning optimizer

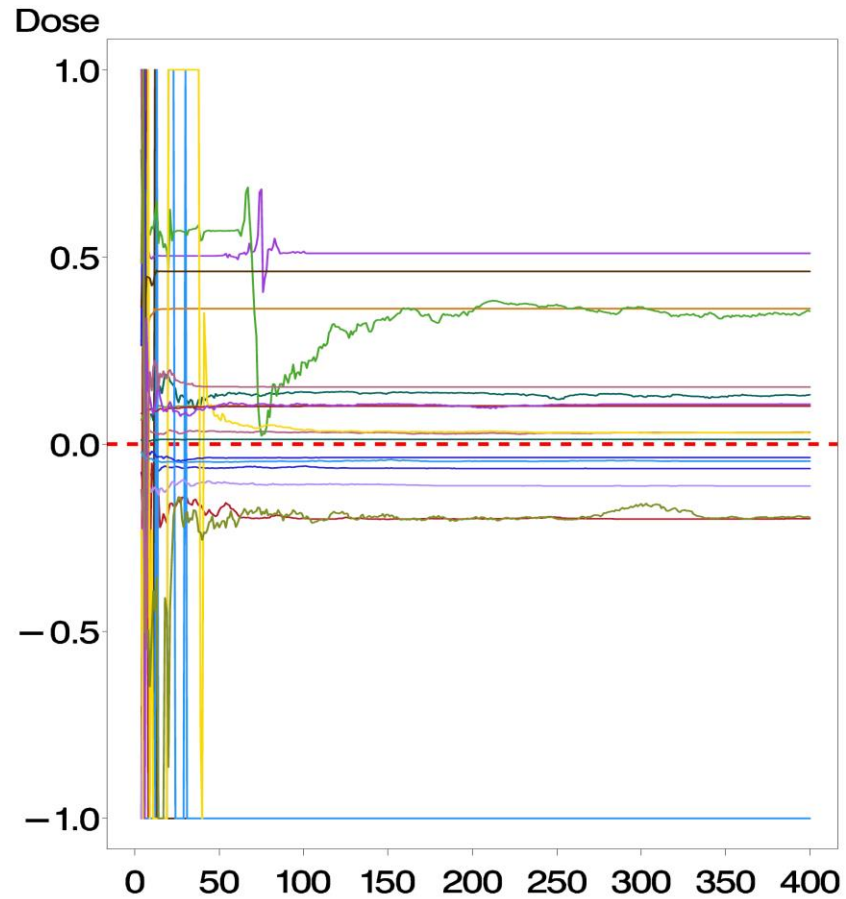
$$y = \eta(x, \theta) + \varepsilon$$

$$\eta(x, \theta) = \theta_1 + \theta_2 x + \theta_3 x^2$$

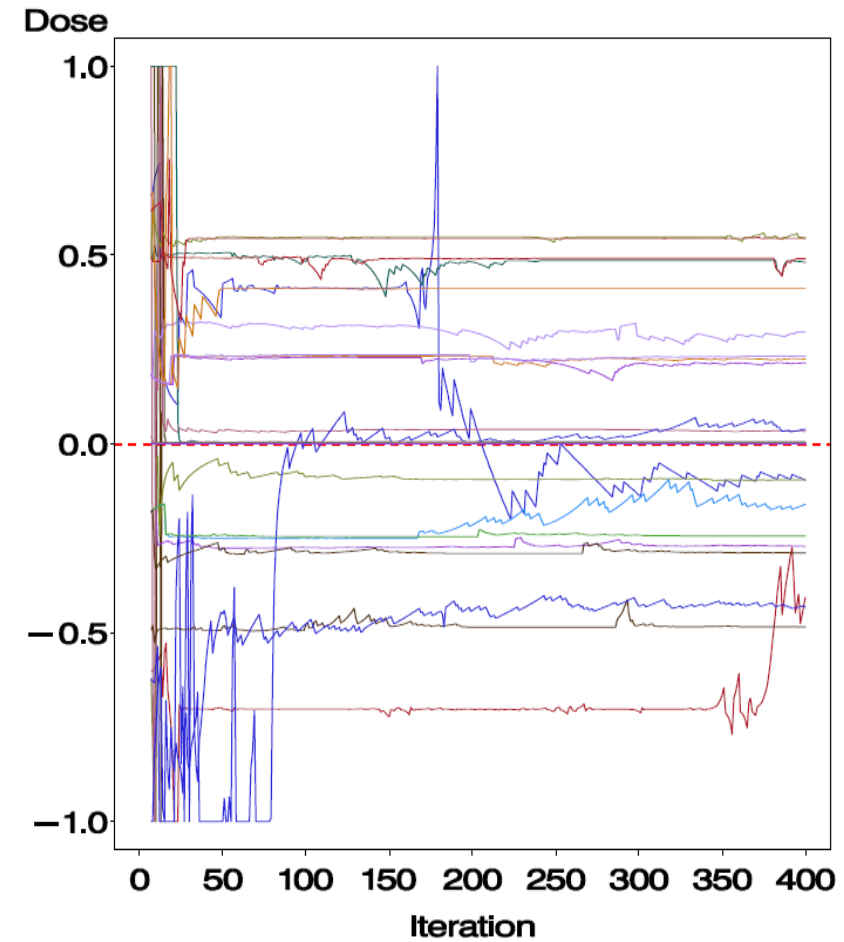
Best intention design: allocate the next patient to the “best” dose

$$x_{n+1} = -\frac{\hat{\theta}_{2,n}}{2\hat{\theta}_{3,n}}$$

A few BID trajectories for Type II problem



Quadratic regression



Binary logistic model with quadratic link

Example 4. Design of large safety trials

Example: Establishing vaccine safety in studies with rare events, Rotarix™ Vaccine Safety Trial

- Public Health Impact of Rotavirus Disease

Nearly all children are infected with RV by age 5 years

The major cause (45%) of severe diarrhea

- U.S.

55 000 hospitalizations/yr

20 to 40 deaths/yr

- Worldwide

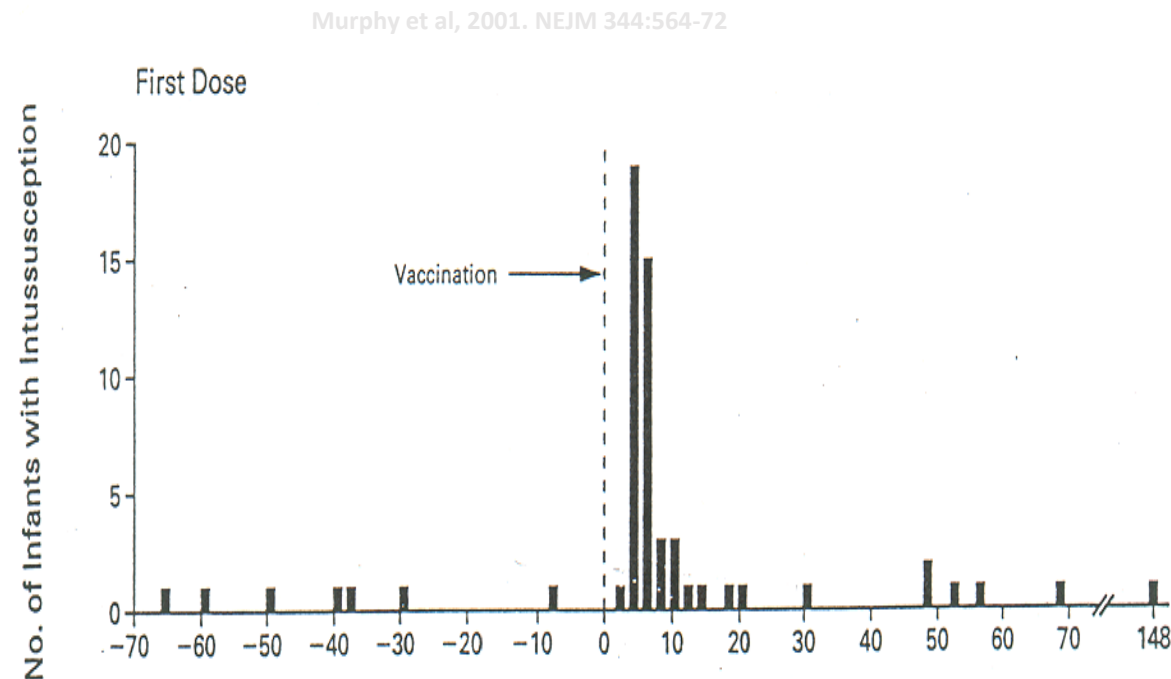
140 million. cases

26% hospitalizations

452 000 deaths/yr

Intussusception (IS) with Rotasheild[®]

- Rotasheild[®] RRV-TV vaccine licensed by Wyeth-Lederle in US 1998, Europe 1999
- In 1999 withdrawn due to increased risk of IS,



• Most common in children between 3 months and 3 years old, but can occur at any age.

Conventional Vaccine Safety Trial

- Determine the safety of Rotarix™ with respect to IS
- Probabilities of IS: p_1 placebo and p_2 vaccine
- Risk Ratio: $RR = p_2/p_1$
- Rule out the null hypothesis

$$H_0 : RR \geq 4$$

- The consumer risk

$$\alpha = \Pr (\text{Accept} \mid RR=4) \leq 0.05$$

- The producer risk

$$\beta = \Pr (\text{Reject} \mid RR=1) \leq 0.1$$

- Assumptions: $p_1 = 3/10\,000$

Conventional (based on “normal” approximation) designs for $\alpha = 0.05$, $\beta = 0.1$, $p_1 = 0.0003$

$\mathcal{R}_*$	Equal randomization			Optimal randomization ratio							
				Minimizing N				Minimizing $E(T_0)$			
	N	$E(T_0)$	$E(T_1)$	h_N	N	$E(T_0)$	$E(T_1)$	h_T	N	$E(T_0)$	$E(T_1)$
30	20782	97	6	3.10	16878	41	5	25.75	30090	19	9
25	22092	86	7	2.95	18189	39	5	21.60	31052	19	9
20	24016	76	7	2.75	20102	37	6	17.35	32451	20	10
15	27186	65	8	2.55	23238	34	7	13.20	35029	21	11
10	33692	56	10	2.25	29644	34	9	8.90	40450	23	12
5.0	57374	52	17	1.85	52975	38	16	4.55	62058	32	19
4.5	63956	53	19	1.80	59483	40	18	4.10	68253	35	20
4.0	73112	55	22	1.75	68551	43	21	3.70	77287	38	23
3.5	86698	59	26	1.65	82025	48	25	3.25	90431	43	27
3.0	108824	65	33	1.55	104022	56	31	2.80	112057	51	34
2.5	150528	79	45	1.45	145566	70	44	2.35	153190	67	46
2.0	252514	114	76	1.35	247360	106	74	1.90	254527	103	76

Design based on a conditional test

- Observed adverse cases: X_1 – on placebo, X_2 – on vaccine plus
- Given $T = X_1 + X_2$ the conditional distribution of X_2 is binomial with parameters (π, T)

$$\pi = \lambda_2 / (\lambda_1 + \lambda_2) = n_2 p_2 / (n_1 p_1 + n_2 p_2) = RR / (h + RR)$$

- Now

$$H_0: \pi \geq \pi^* \quad \text{versus} \quad H_1: \pi < \pi^*$$

- UMP test: reject H_0 if $X_2 \leq C_\alpha$,

$$P(X_2 \leq C_\alpha \mid \pi \geq \pi^*) \leq \alpha$$

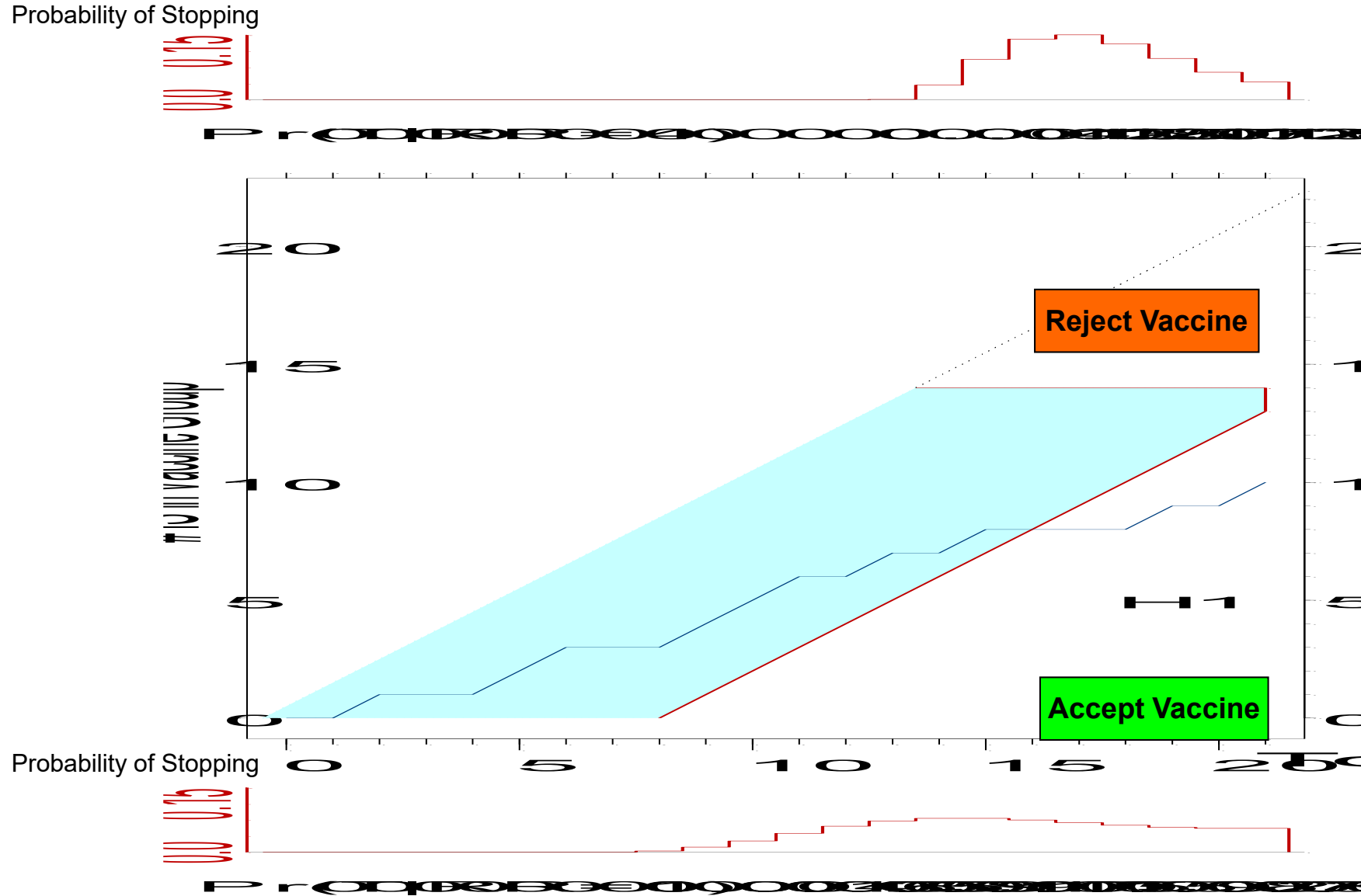
- Waiting time and sample size are random variables

Lehmann E.L. Testing statistical hypotheses, Wiley, 1959, Ch. 4.5

Number of events and critical values for a specified consumer risk $\alpha=0.05$ and randomization ratio $h=1$

$\mathcal{R}_*$	$\beta = 0.05$		$\beta = 0.1$		$\beta = 0.2$	
	T^*	C_α	T^*	C_α	T^*	C_α
5.00	21	14	19	12	12	7
4.50	23	15	19	12	15	9
4.00	28	18	21	13	17	10
3.50	33	21	26	16	19	11
3.00	42	26	33	20	26	15
2.50	58	35	46	27	32	18
2.00	98	57	79	45	58	32

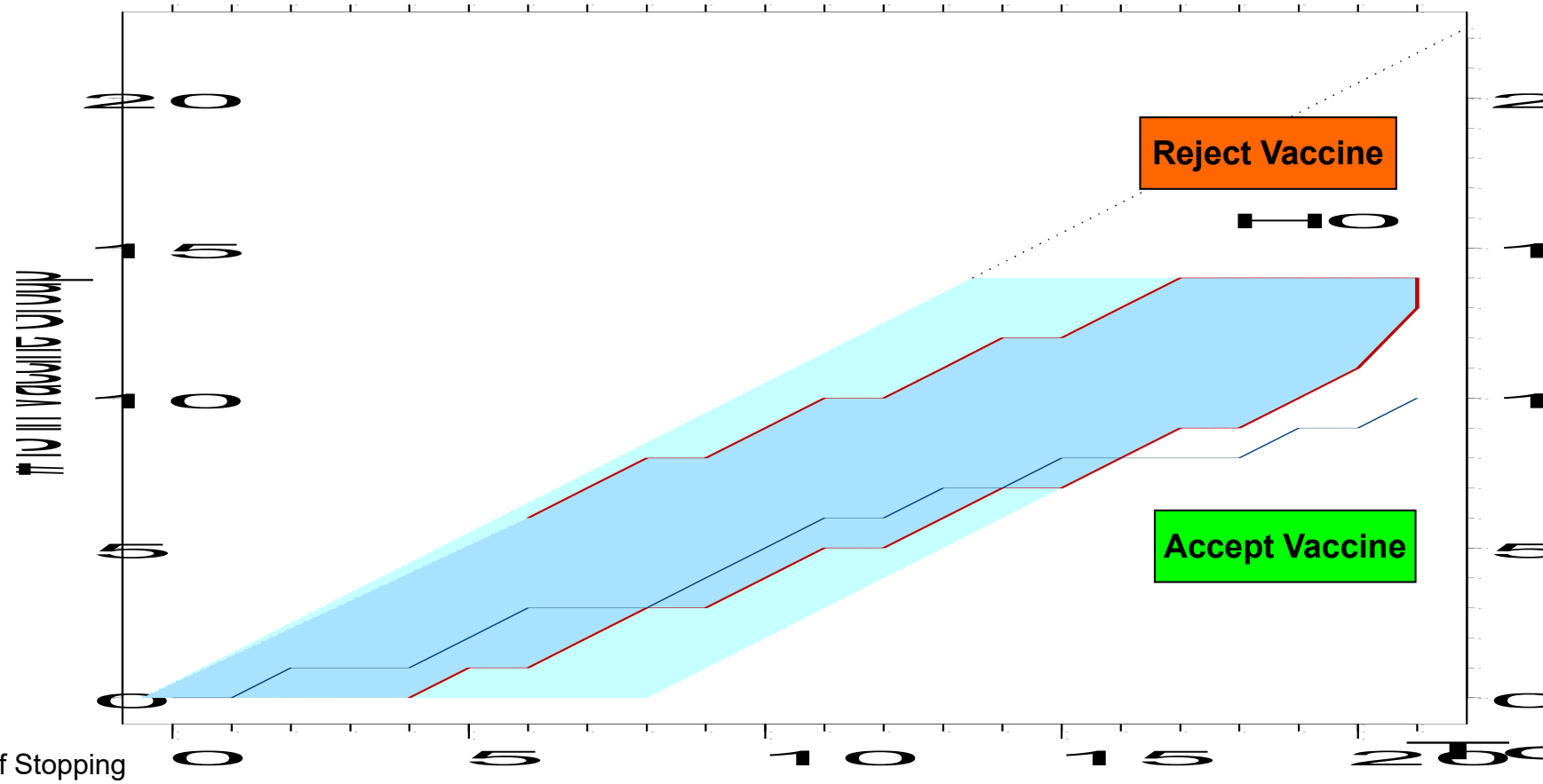
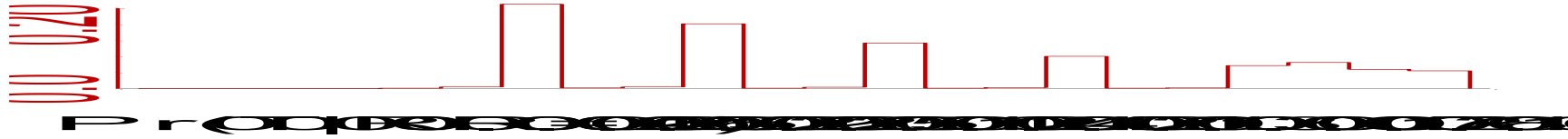
Curtailed Design



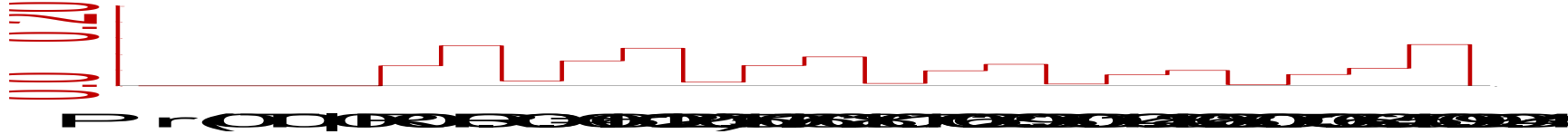
The process halts as soon as the decision is clear, based on the current data

Sequential Design

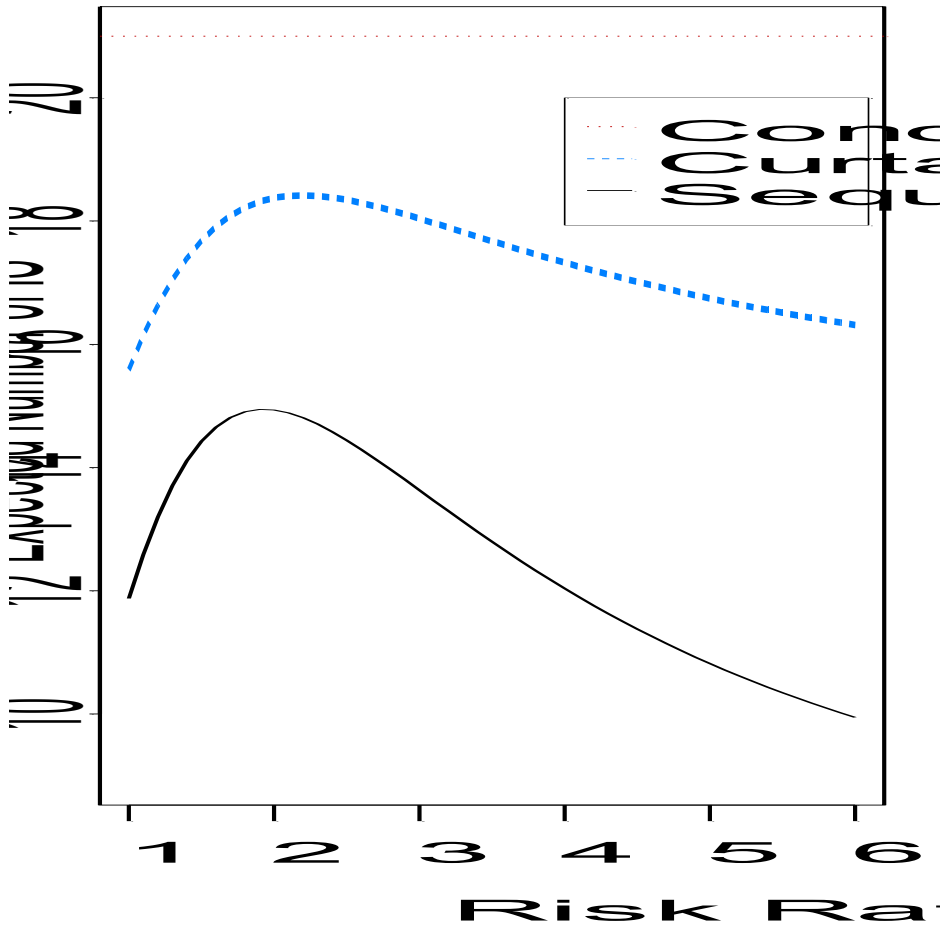
Probability of Stopping



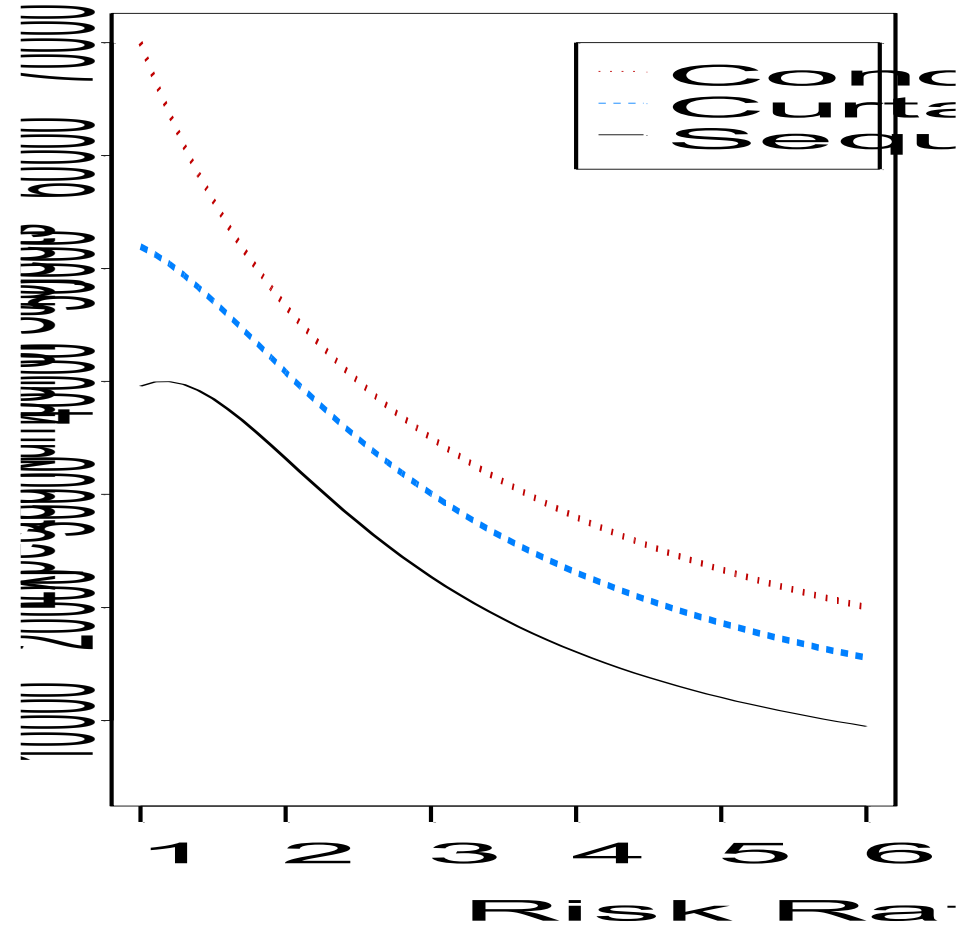
Probability of Stopping



Spare your Patients



Spare your Money



A few comments

- Conventional methods based on normal approximation are inappropriate in safety trials with rare events
- Conditional test based on Poisson approximation is a good alternative
- The major drawback: it may require large sample sizes when the rates are low
- Curtailment is a reasonable enhancement of the procedure
- Sequential design provides a further considerable improvement
- This method was used for the design of clinical trials for the J&J COVID-19 vaccine

Example 5. Structured meta trials: 6 Cancer Types and 5 Drugs

Operational constraints for blocks (sub trials):

- No more than 3 cancer types per block
- Only 2 drugs per block

	Cancer					
Block	C1	C2	C3	C4	C5	C6
1	D1,5	D1,5	D1,5			
2	D1,2				D1,2	D1,2
3	D2,3		D2,3	D2,3		
4	D3,4	D3,4				D3,4
5	D4,5			D4,5	D4,5	
6		D1,3		D1,3	D1,3	
7		D2,4	D2,4		D2,4	
8			D3,5		D3,5	D3,5
9			D1,4	D1,4		D1,4
10		D2,5		D2,5		D2,5

Properties: • Every pair of drugs at one trial • Every pair of cancer types at two trials • Every drug with every cancer type at two trials	Benchmarking: in reality “practical” designs take into account medical knowledge, disease prevalence, differing enrollment rates per cancer type and competing products
---	---

Combinatorial design *

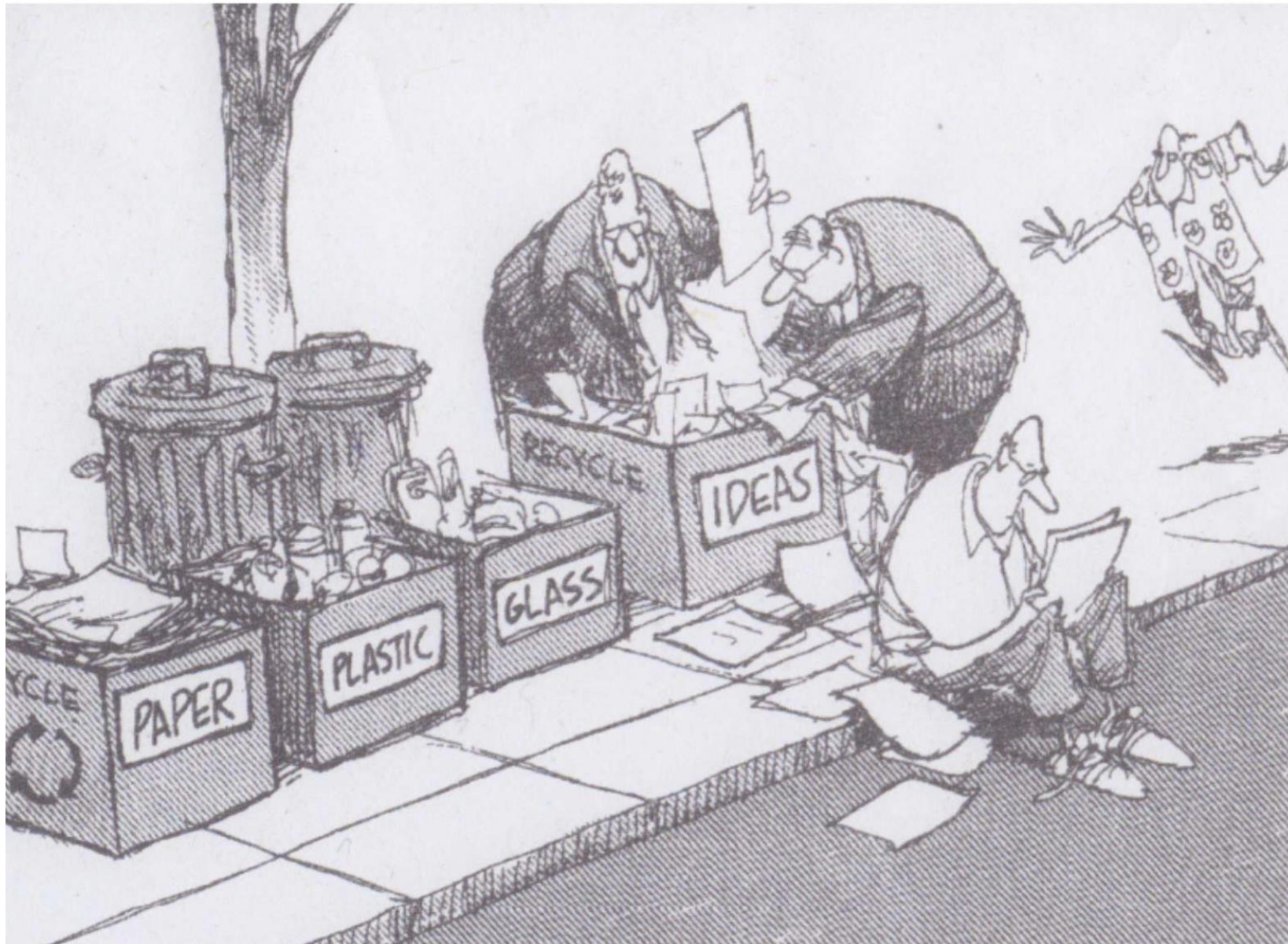
Latin square	C1	C2	C3	C4	C5	C6
B1	T1	T2	T3	T4	T5	T6
B2	T2	T1	T6	T5	T4	T3
B3	T3	T4	T1	T2	T6	T5
B4	T4	T6	T5	T1	T3	T2
B5	T5	T3	T2	T6	T1	T4
B6	T6	T5	T4	T3	T2	T1

Efficacy for the i -th subject having C_k and B_l , and treated with T_j :

$$Y_{ijkl} = \mu + \tau_j + \gamma_k + \beta_l + \varepsilon_{ijkl}$$

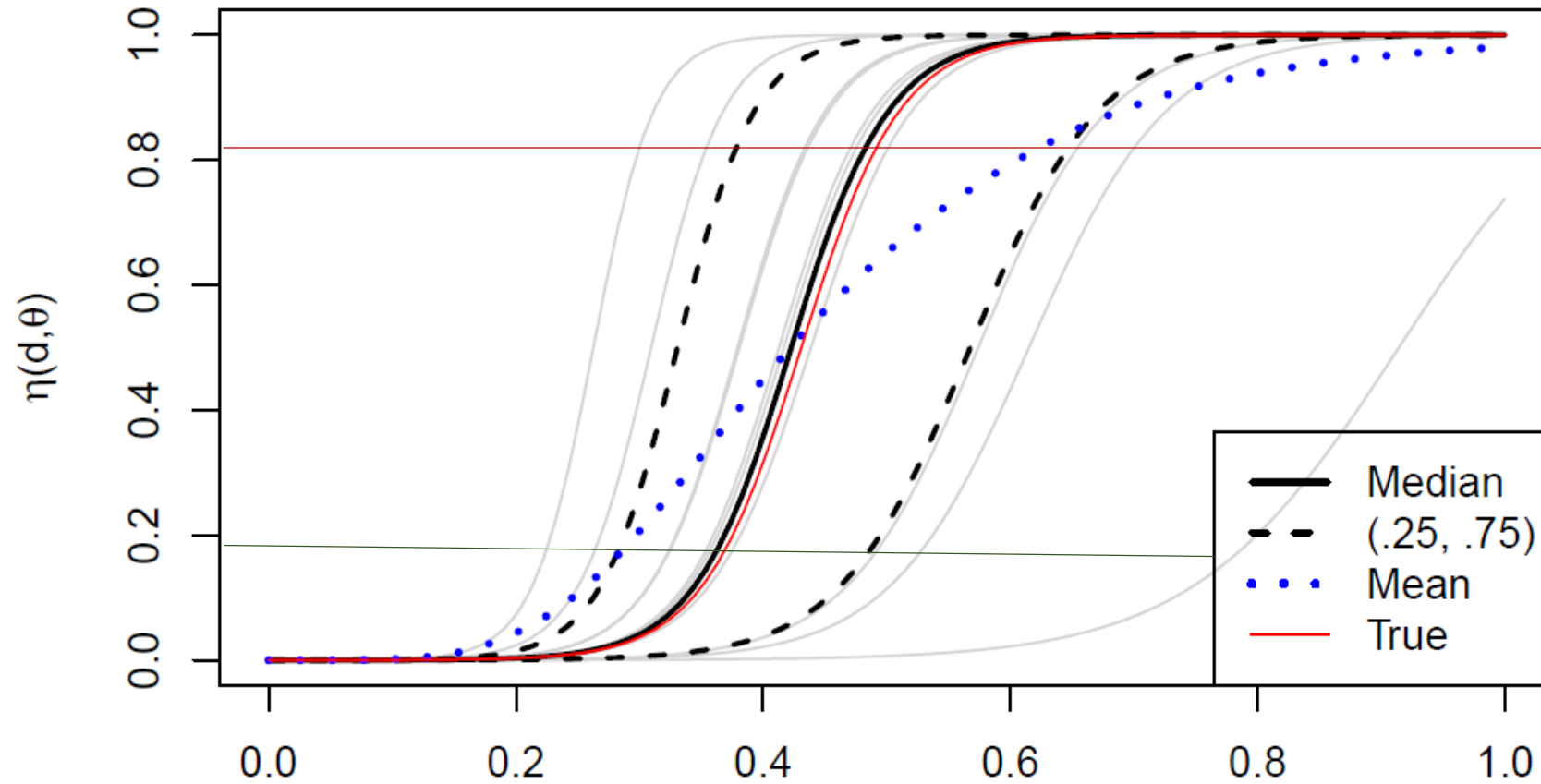
* Too good to be used in pharmaceutical practice, they cannot compete in popularity with: Basket trials, Umbrella trials, Platform trials, Master protocols, etc..

Sometimes it helps



Complimentary Slides

Location of most informative doses



Support points of locally D-optimal designs are points that map 0.18 and 0.82 quantiles

Distribution	Density	Mean Variance	Information
Bernoulli(p) $0 \leq p \leq 1$	$p^y(1-p)^{1-y}$	p $p(1-p)$	$1/[p(1-p)]$
Geometric(p) $0 \leq p \leq 1$	$(1-p)^yp$	$(1-p)/p$ $(1-p)/p^2$	$1/[p^2(1-p)]$
Binomial(p, n) $0 \leq p \leq 1$	$\binom{n}{y}p^y(1-p)^{n-y}$	np $np(1-p)$	$n/[p(1-p)]$
Neg. Bin. (p, m) $0 \leq p \leq 1$	$\binom{m+y-1}{m-1}p^m(1-p)^y$	$m(1-p)/p$ $m(1-p)/p^2$	$m/[p^2(1-p)]$
Hypergeometric (p, N, n) $0 \leq p \leq 1$	$\frac{\binom{n}{y}\binom{N-n}{Np-y}}{\binom{N}{Np}}$	np $np(1-p)(N-n)/(N-1)$	$\frac{(N-1)n}{p(1-p)(N-n)}$
Poisson(λ) $\lambda > 0$	$\frac{\lambda^ye^{-\lambda}}{y!}$	λ λ	$1/\lambda$

Distribution	Density	Mean Variance	Support	Information matrix
Normal(a, σ^2) $-\infty < a < \infty$ $\sigma > 0$	$\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y-a)^2}{2\sigma^2}}$	a σ^2	$-\infty, \infty$	$\sigma^{-2} \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{2}\sigma^{-2} \end{pmatrix}$
Beta(α, β) $\alpha > 0$ $\beta > 0$	$B(\alpha, \beta)y^{\alpha-1}(1-y)^{\beta-1}$	$\frac{\alpha/(\alpha+\beta)}{\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}}$	$0 < y < 1$	$\begin{pmatrix} \psi'(\alpha) - \psi'(\alpha+\beta) & -\psi'(\alpha+\beta) \\ -\psi'(\alpha+\beta) & \psi'(\beta) - \psi'(\alpha+\beta) \end{pmatrix}$
Gamma(α, β) $\alpha > 0$ $\beta > 0$	$\frac{1}{\Gamma(\alpha)\beta^\alpha} y^{\alpha-1}e^{-y/\beta}$	$\frac{\alpha\beta}{\alpha\beta^2}$	$0, \infty$	$\begin{pmatrix} \psi'(\alpha) & 1/\beta \\ 1/\beta & \alpha/\beta^2 \end{pmatrix}$
Logistic(a, b) $-\infty < a < \infty$ $b > 0$	$\frac{e^{-(y-a)/b}}{b[1 + e^{-(y-a)/b}]^2}$	$\frac{a}{b^2\pi^2/3}$	$-\infty, \infty$	$\frac{1}{3} \begin{pmatrix} b^{-2} & 0 \\ 0 & 1 + b^{-2} \end{pmatrix}$
Cauchy(a, b) $-\infty < a < \infty$ $b > 0$	$\frac{b}{\pi[b^2 + (y-a)^2]}$	Do not exist	$(-\infty, \infty)$	$\frac{1}{2b^2}I$
Weibull(α, β) $\alpha > 0$ $\beta > 0$	$\frac{\alpha}{\beta}(\frac{y}{\beta})^{\alpha-1}e^{-(y/\beta)^\alpha}$	$\frac{\beta\Gamma(1+\alpha^{-1})}{\beta^2[\Gamma(1+2\alpha^{-1}) - \Gamma^2(1+\alpha^{-1})]}$	$0, \infty$	$\begin{pmatrix} \frac{\pi^2}{6} + \frac{(1-\gamma)^2}{\alpha^2} & \frac{\gamma-1}{\beta} \\ \frac{\gamma-1}{\beta} & \frac{\alpha^2}{\beta^2} \end{pmatrix}$
Pareto(α, σ) $\alpha > 2$ (for variance) $\sigma > 0$	$\frac{\alpha}{x}(\frac{y}{\sigma})^{-\alpha}$	$\sigma \left(\frac{\alpha}{\alpha-1} \right)$ $\frac{\alpha\sigma^2}{(\alpha-1)^2(\alpha-2)}$	$0, \infty$	$\begin{pmatrix} \frac{\alpha}{\sigma^2(\alpha+2)} & -\frac{1}{\sigma(\alpha+1)} \\ -\frac{1}{\sigma(\alpha+1)} & \frac{1}{\alpha^2} \end{pmatrix}$
Double exponential (Laplace)(a, b) $-\infty < a < \infty$ $b > 0$	$\frac{e^{- y-a /b}}{2b}$	$\frac{a}{2/b^2}$	$-\infty, \infty$	$\frac{1}{b^2}I$

$\psi(\alpha) = \frac{\Gamma'(\alpha)}{\Gamma(\alpha)}$ and $\psi'(\alpha) = \frac{d\psi(\alpha)}{d\alpha}$ are the digamma and trigamma functions; $\gamma = 0.5772$ is Euler's constant,

Distribution	New parameters	Information matrix
$N(a, \sigma^2)$	$\vartheta_1 = a$ $\vartheta_2 = \ln \sigma^2$	$\begin{pmatrix} 1/e^{\vartheta_2} & 0 \\ 0 & 1/2 \end{pmatrix}$
$N(a, a^2)$	$\vartheta = \ln a$	3
$N(a, k^2 a^2)$	$\vartheta_1 = \ln a$ $\vartheta_2 = k^2$	$\begin{pmatrix} 2 + 1/\vartheta_2 & 1/\vartheta_2 \\ 1/\vartheta_2 & 1/(2\vartheta_2^2) \end{pmatrix}$
...	...	...
Cauchy(a, b)	$\vartheta_1 = a$ $\vartheta_2 = \ln b$	$\frac{1}{2} \begin{pmatrix} e^{-2\vartheta_2} & 0 \\ 0 & 1 \end{pmatrix}$
Weibull(α, β)	$\vartheta_1 = \ln \alpha$ $\vartheta_2 = \ln \beta$	$\begin{pmatrix} \frac{\pi^2}{6} + (1 - \gamma)^2 & \frac{\gamma-1}{e^{\vartheta_2}} \\ \frac{\gamma-1}{e^{\vartheta_2}} & e^{2\vartheta_1} \end{pmatrix}$

$$\mu(\vartheta) = n \begin{pmatrix} \vartheta_1 & 0 & \dots & 0 \\ 0 & \vartheta_2 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & \vartheta_l \end{pmatrix}^{-1} + \frac{n}{\vartheta_{l+1}} ll^T$$